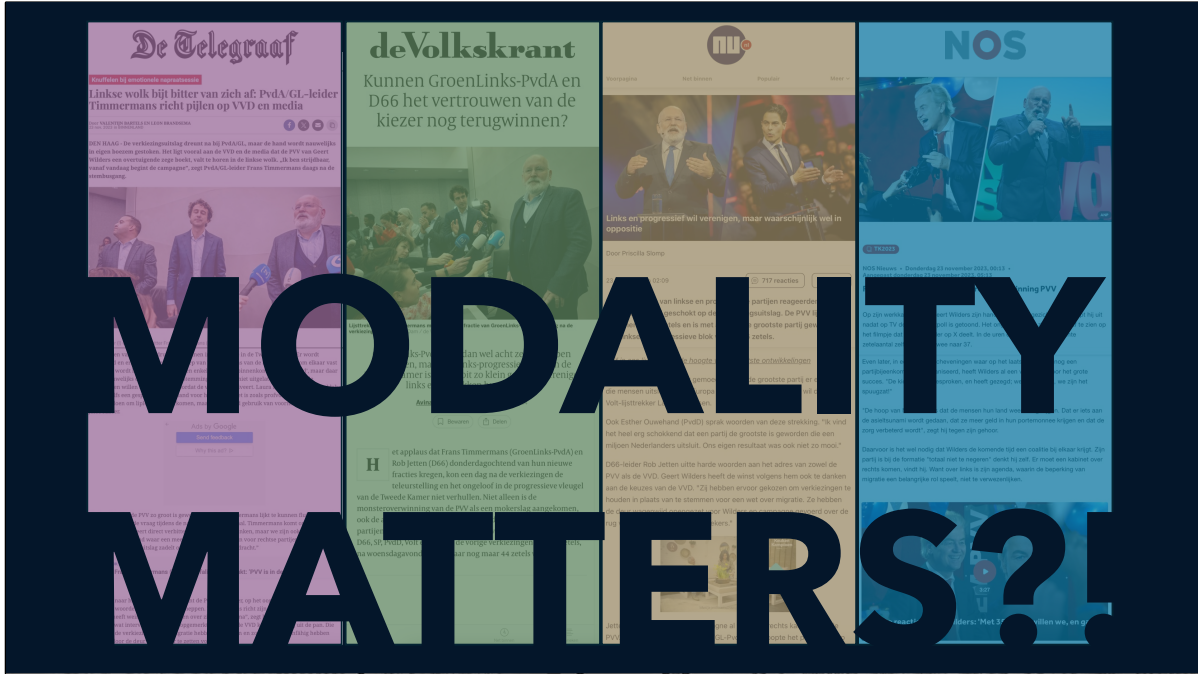




This is a recording of the presentation I gave at the ECREA interim conference in Innsbruck, September 5 2025.

How Multi-Modal Media Prominence Drives Political Recognition and Party Support



Does modality matter for media effects of political news coverage? Do we remember politicians better when they appear in both text and image, compared to when they appear in either of them, that is, in monomodal form? This is the central question for this project and I am excited to walk through it together.

The Netherlands - November 23, 2023 'The Day After'



Let's go back to what we in the Netherlands considered the "day after"; November 23 in 2023. After a long campaign, the election results made clear that PVV, the populist and anti-islam party for freedom of blonde-haired Geert Wilders, won these elections by a landslide, leaving mainstream-right and left-wing parties stunned.



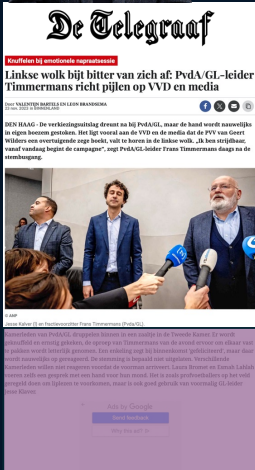
*"Left-wing cloud bites back bitterly:
PvdA/GL leader Timmermans takes aim at
VVD and media"*

Title: mocking, blaming

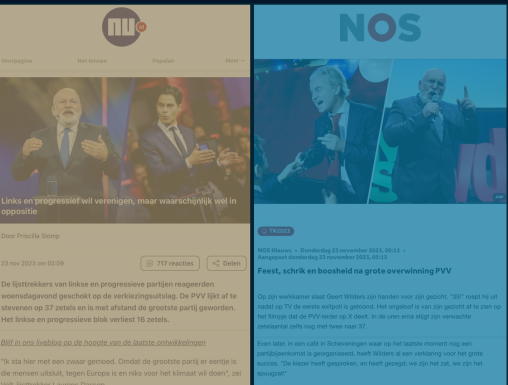
Image: messy, beaten

The Netherlands - November 23, 2023

'The Day After'







Interestingly, a lot of coverage focused on the losers of this election more than the winner. In particular, there was a special spotlight reserved for the green-social party leader Frans Timmermans. Frans Timmermans, for context, is a former Europarlementarian who had also served several roles within Dutch government in the past. He may rightfully be called the “establishment” from the left side of the political spectrum.

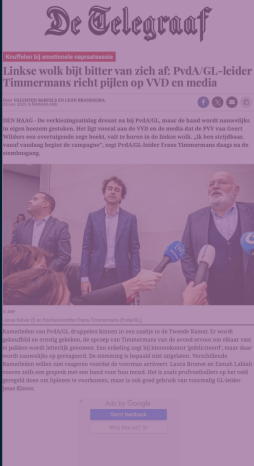
De Telegraaf is a tabloid style newspaper and does not particularly favor anything leftist. This is confirmed in their news framing. We see, here, an image where the party prominent figures of the green-social party, including Timmermans, appear wrecked or “beaten”. The title is mocking and portrays Timmermans as someone who looks to blame others instead of engaging with self-reflection.



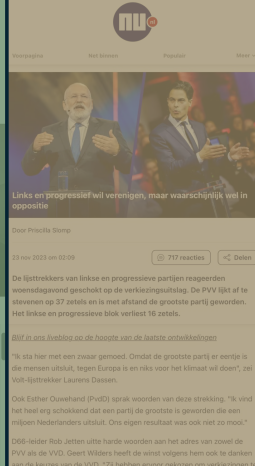
"Festivity, scare and anger after PVV's big victory"
Title: descriptive, exemplary
Image: furious

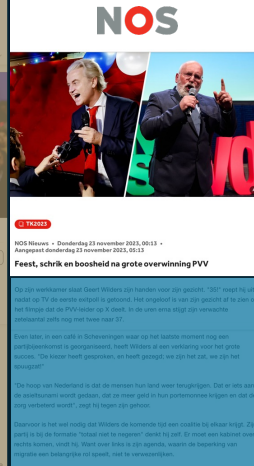
The Netherlands - November 23, 2023

'The Day After'









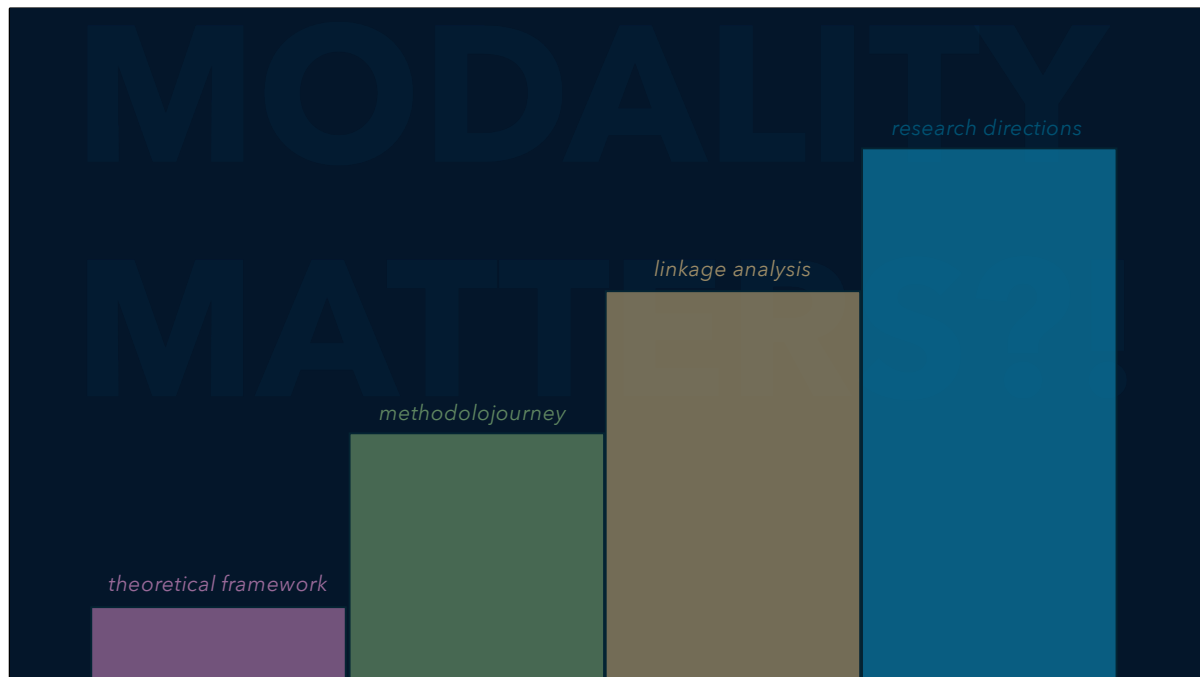
Last, there is the NOS, the public broadcaster. They created a seemingly conflicted frame where they contrast winner Geert Wilders and loser Frans Timmermans, yet depicting Timmermans as furious and combatant.

which (combination of) modalities
in political news coverage
are most influential
for a politician's perceived prominence
by the public?

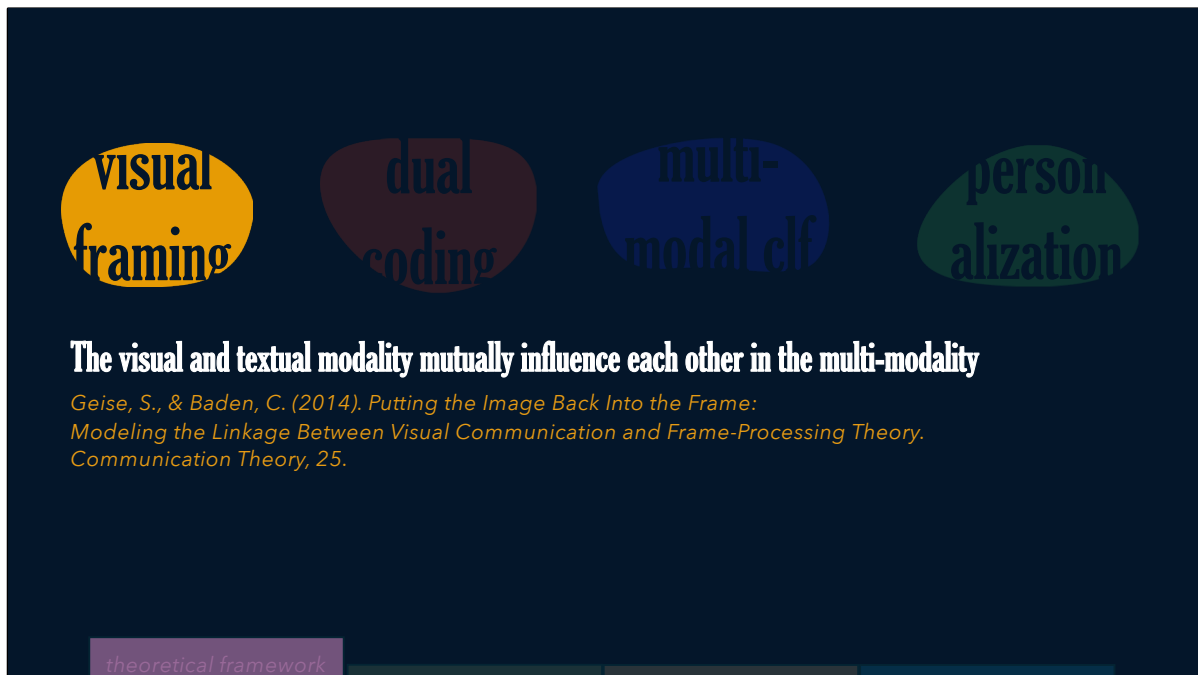


What article you encountered on November 23, might have had a strong influence on what you associated Frans Timmermans with – a loser or a fighter. We wonder what modality is most influential in establishing this framing effect. But to get there, we first want to understand what modality is most influential for recalling a politician in the first place. We therefore question: **which (combination of)**

in
for a **perceived** **are most influential**
by the **?**



This presentation is structured as follows; we first walk through the theoretical framework, then discuss the methods, or methodoljourney as I'd like to call it, the presentation of results, followed by some questions to spark a fruitful discussion



To start of with our theoretical base.

First, we deepened our understanding of visual framing. Most importantly, following Geise and Bades, we do not see text and image as separate, independent dimensions of communication. They interact and complement. We encounter them on news websites, printed papers, and on social media in combination – so why study them in isolation? That even rhymes, just to say.

visual
framing

dual
coding

multi-
modal clf

person
alization

Multi-modal information enhances information processing and recall

Clark, J., & Paivio, A. (1991). Dual Coding Theory and Education.

Educational Psychology Review, 3, 149-210. <https://doi.org/10.1007/BF01320076>

theoretical framework

We also assume, from Dual Coding theory, that multimodal information enhances information processing and recall. This is an old theory however, which we would love to see reviewed, but for now, it is the strongest scientific reference to build our hypothesis on.

visual
framing

dual
coding

multi-
modal clf

person
alization

News images and their headlines are not necessarily aligned in terms of semantics

Oostdijk, N., van Halteren, H., Başar, E., & Larson, M. (2020). The Connection between the Text and Images of News Articles: New Insights for Multimedia Analysis. Twelfth International Conference on Language Resources and Evaluation, Marseille, France.

theoretical framework

We combine a politician's textual coverage with their visual coverage per article, for which we look at the modality congruence. However, that need not to imply semantic congruence. It was found that news images and headlines do not form unique, 1:1 relationships – a news image can fit multiple headlines, and one headline can apply to multiple images to still be deemed fitting by news audiences.

visual
framing

dual
coding

multi-
modal ctf

person
alization

The shift in attention from political issues to political personalities

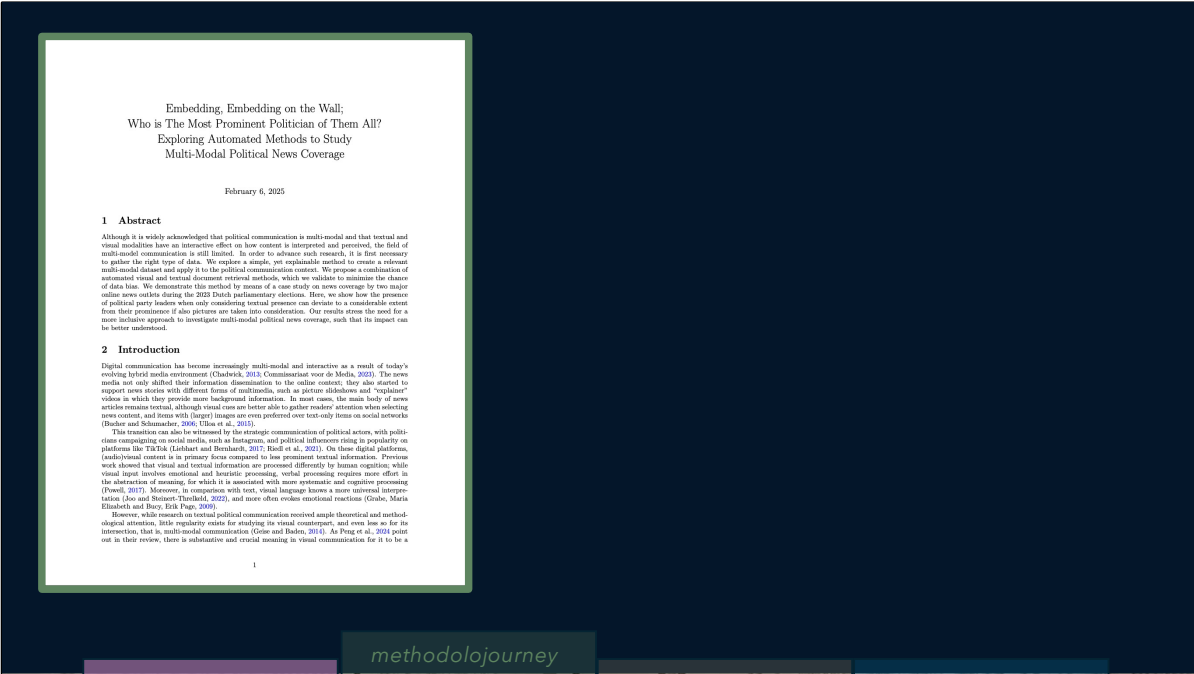
Boumans, J. W., Boomgaarden, H. G., & Vliegenthart, R. (2013). Media Personalisation in Context: A Cross-National Comparison between the UK and the Netherlands, 1992-2007. Political Studies, 61(S1), 198-216. <https://doi.org/10.1111/1467-9248.12013>

theoretical framework

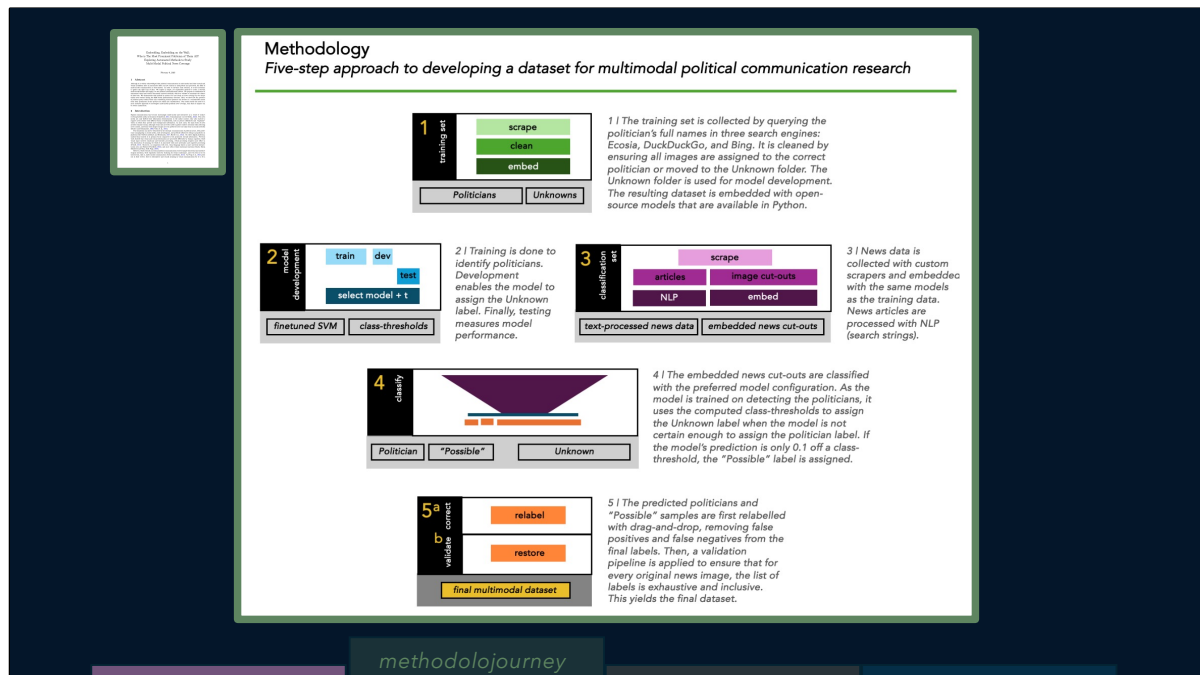
Last, we place our study in the field of personalization of politics. The Personalization hypothesis states that over the last decade, the political information we consume has grown predominantly centered around the politician more than parties, issues, or institutions. This angle is relevant for us, as we will zoom in on the party leaders exclusively, and wonder whether potential effects differ between them.



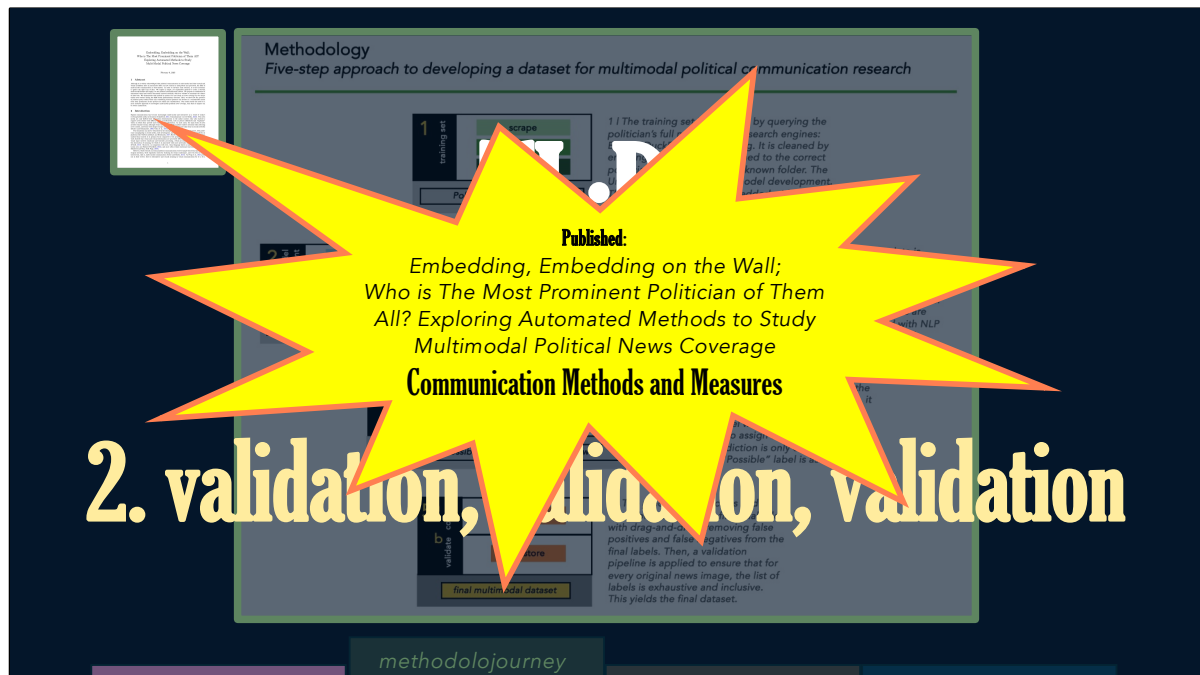
Now, it's time for my favorite part of this job: methodology.



Collecting multimodal datasets is something I have been doing for a while now.

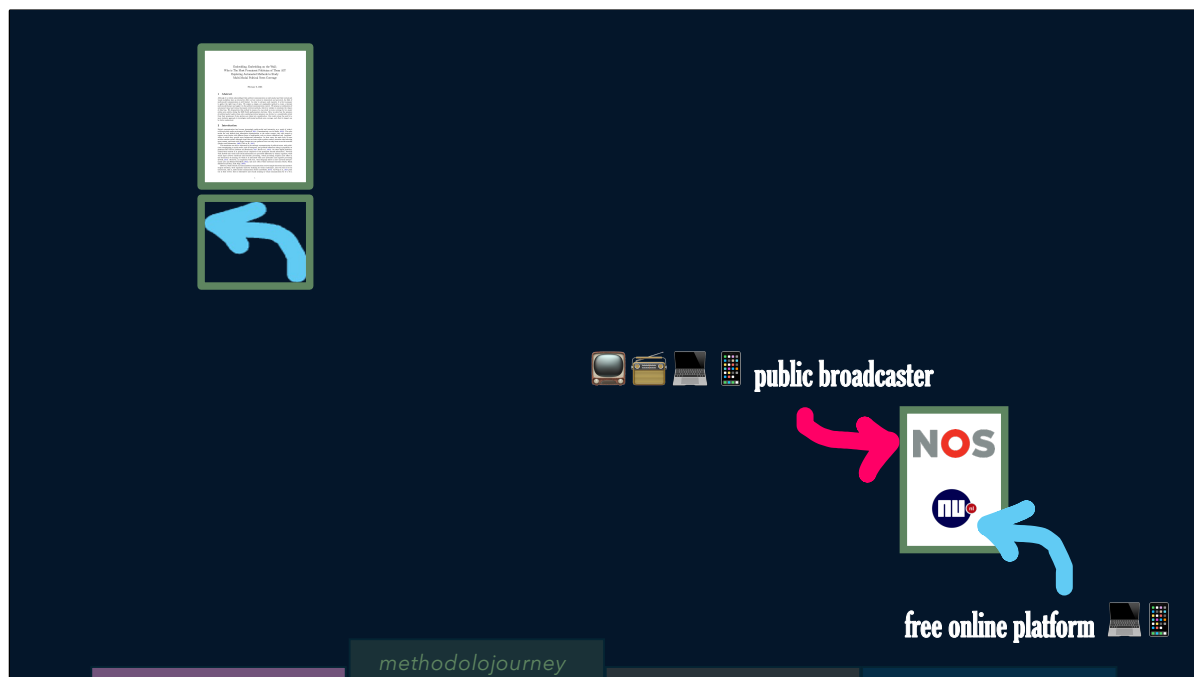


I spent the first year and a half of my PhD creating an explainable and open-source-reliant pipeline to detect politicians' visual appearances in news images.

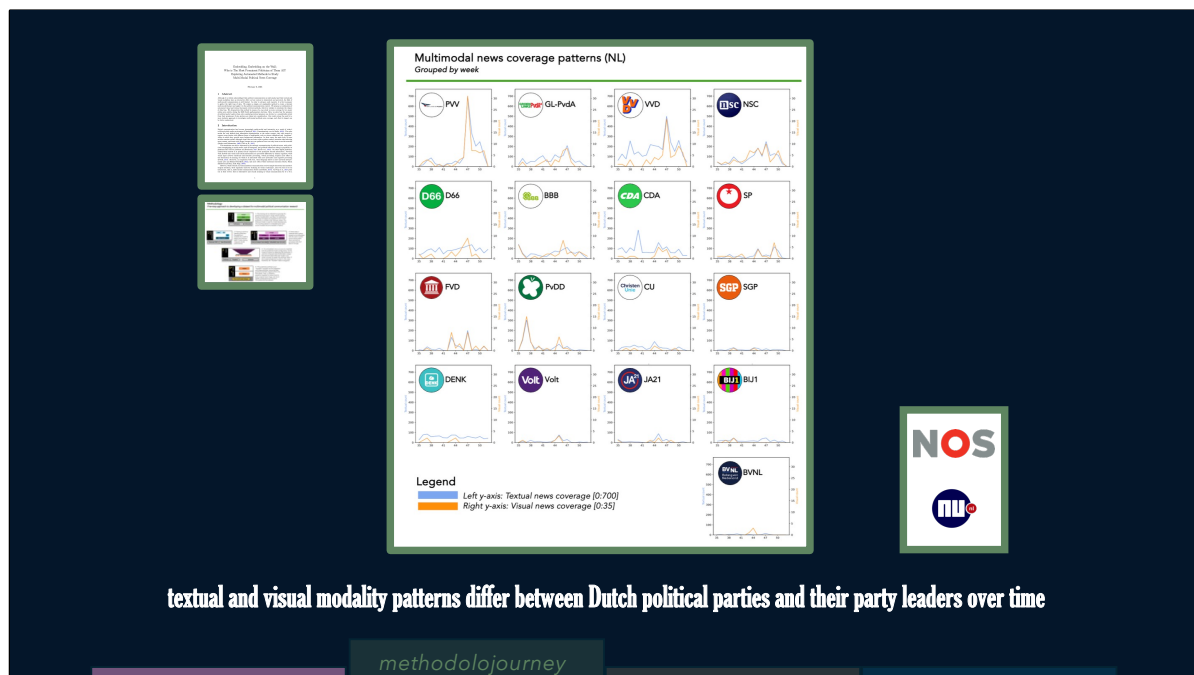


TL;DR - this was a lot of work. And validation, here, is crucial.

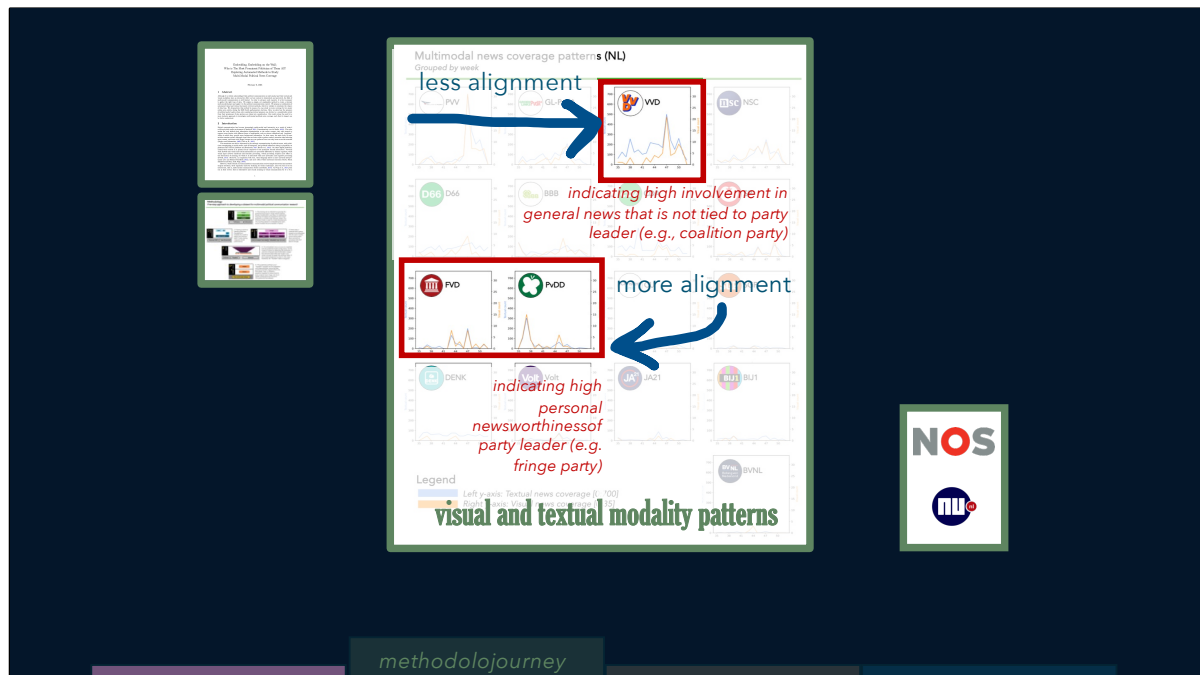
You can read all about it in my publication. For now, the most important is that this pipeline allowed us to construct a pretty reliable multimodal dataset where we track politicians' visual and textual appearances per article. That enables us to see, per article and per politician, whether their coverage was monomodal (only visual, only textual) or multimodal (both).



I will quickly showcase what modality alignment, when a party leader appears in the image and the text, looks like for Dutch party leaders in two online news outlets



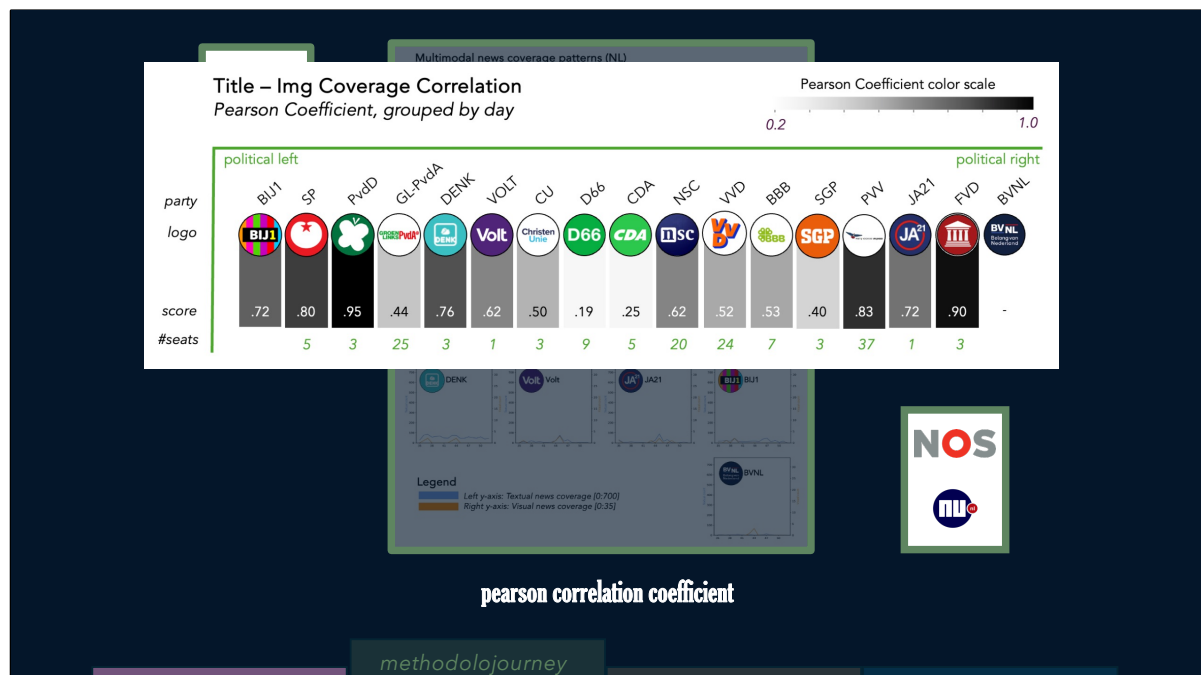
Short answer: this differs greatly per party. And honestly, that is also what we would expect. A few things stand out here.



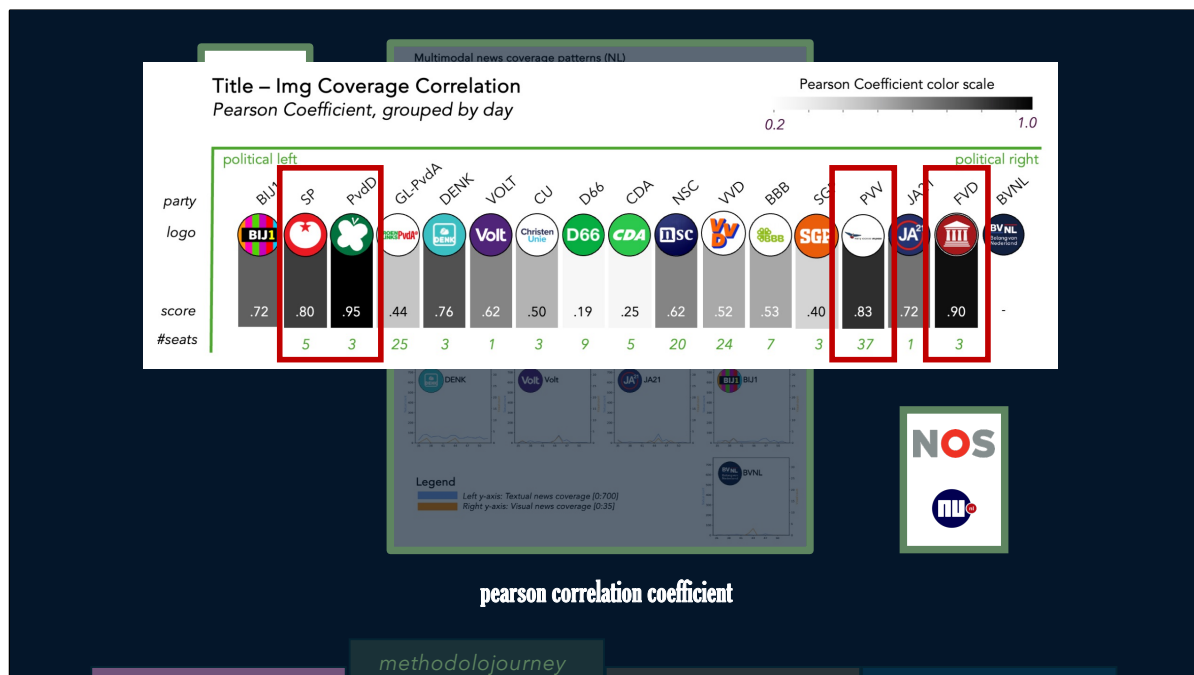
First, we find that there is less modality alignment for coalition parties as their textual coverage is structurally higher than multimodal coverage. This deals with the fact that ministers, junior ministers and state secretaries other than the party leader, often appear visually when the article mentions the party. See it as an accountability game - part of the job for news outlets.

However, for this study we look at party leaders, so these visual appearances of non-party leaders do not count.

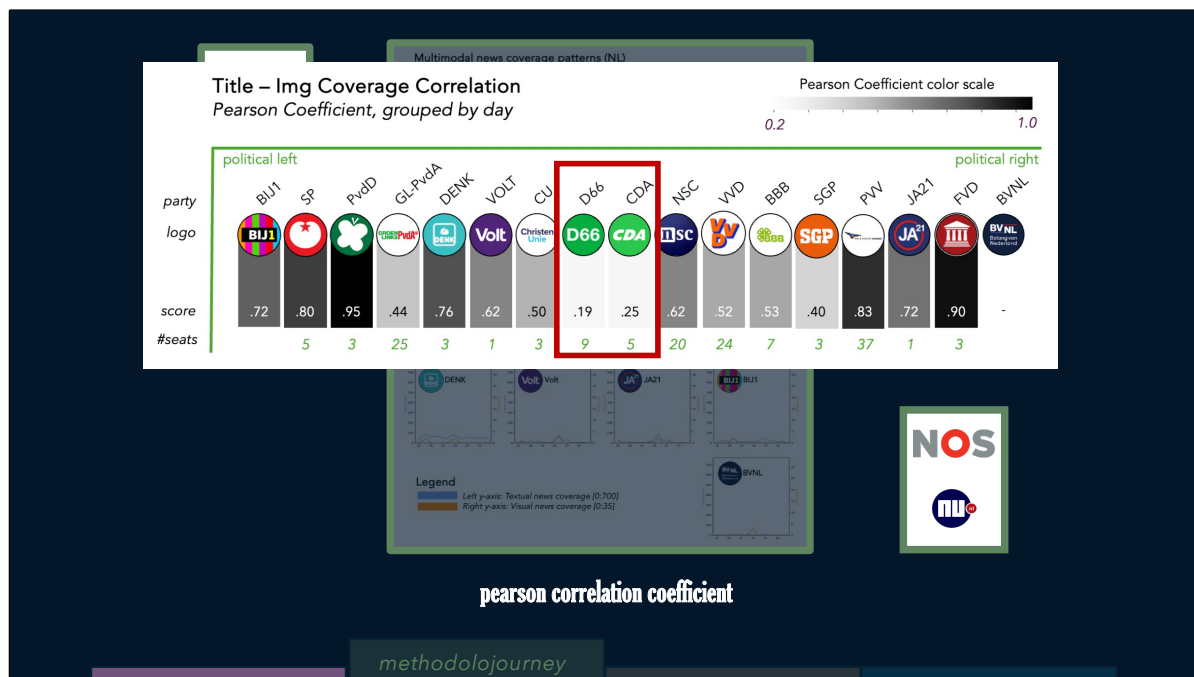
On the other hand, we find that niche parties on the far right and far left edges of the political spectrum do show a relatively high alignment between textual and visual coverage of the party leaders.



This is also reflected in the pearson coefficient for the modality alignment, or title-img coverage correlation, for all political parties.



The more ideologically outspoken and newsworthy a leader is, the more alignment we find, meaning that the news coverage really is focused on the leader and therefore more personalized.



And the reverse is true for “steady state” or mainstream coalition parties.

there's something up — but we must
enrich the data to see more



methodolojourney

But to draw more meaningful conclusions to what this might imply for news audiences, we expanded our multimodal dataset...

there's something up — but we must
enrich the data to see more

 hybrid (traditional/tabloid) newspaper;
considered right on the political spectrum

 hybrid (traditional/quality) newspaper;
considered left on the political spectrum



methodolojourney

... by including two well read news outlets on the left and right of the political spectrum.

**which (combination of) modalities
in political news coverage
are most influential
for a politician's perceived prominence
by the public?**

linkage analysis

Building up to the analysis, let's recall our research question:

**which (combination of) in are most influential for a perceived by
the ?**

**which (combination of) modalities
in political news coverage
are most influential
for a politician's perceived prominence
by the public?**



linkage analysis

We analyse this through a linkage design, which is at the time of writing considered state of the art in closely studying media effects. How does it work?

**which (combination of) modalities
in political news coverage
are most influential
for a politician's perceived prominence
by the public?**



linkage analysis

We take a sample of a population, here: Dutch people. I refer to them as respondents

**which (combination of) modalities
in political news coverage
are most influential
for a politician's perceived prominence
by the public?**



linkage analysis

We ask them about their political perceptions and attitudes, to collect what we call “public opinion data”

**which (combination of) modalities
in political news coverage
are most influential
for a politician's perceived prominence
by the public?**



linkage analysis

And we do so longitudinally, that is, over an extended period of time

**which (combination of) modalities
in political news coverage
are most influential
for a politician's perceived prominence
by the public?**



linkage analysis

In this case, that period included the run-up to election day and its aftermath, consisting of 7 waves

**which (combination of) modalities
in political news coverage
are most influential
for a politician's perceived prominence
by the public?**



linkage analysis

On the other hand, we track political news coverage

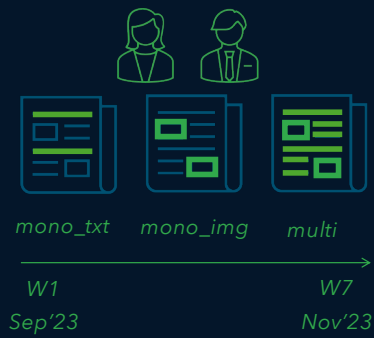
**which (combination of) modalities
in political news coverage
are most influential
for a politician's perceived prominence
by the public?**



linkage analysis

We know, with help of the pipeline that was mentioned earlier, whether politicians and parties appeared only in text, only in the image, or in both

which (combination of) modalities
in political news coverage
are most influential
for a politician's perceived prominence
by the public?



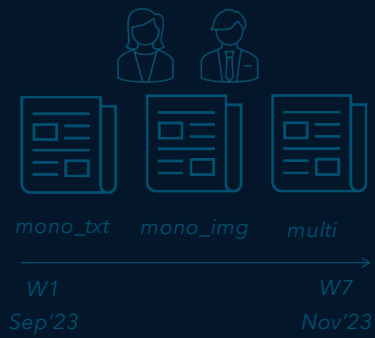
linkage analysis

And we track this monomodal versus multimodal political news coverage over the same time period as we have public opinion data from.

which (combination of) modalities
in political news coverage
are most influential
for a politician's perceived prominence
by the public?



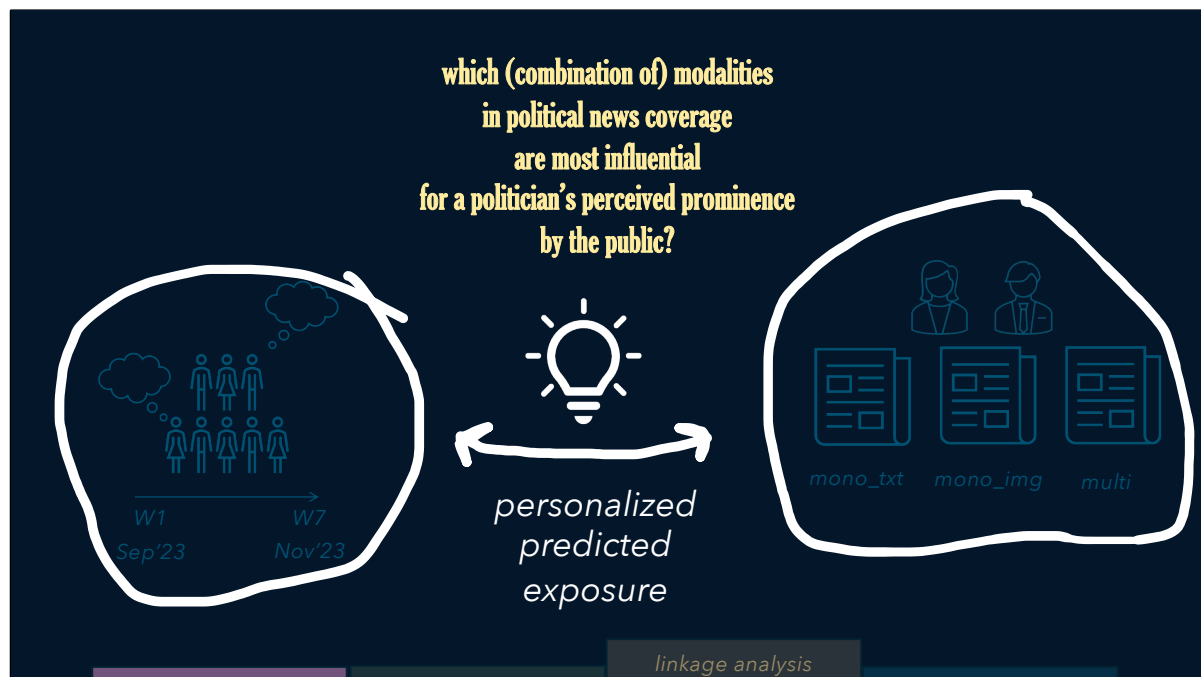
...?



linkage analysis

So...

...What crucial information am I missing here?



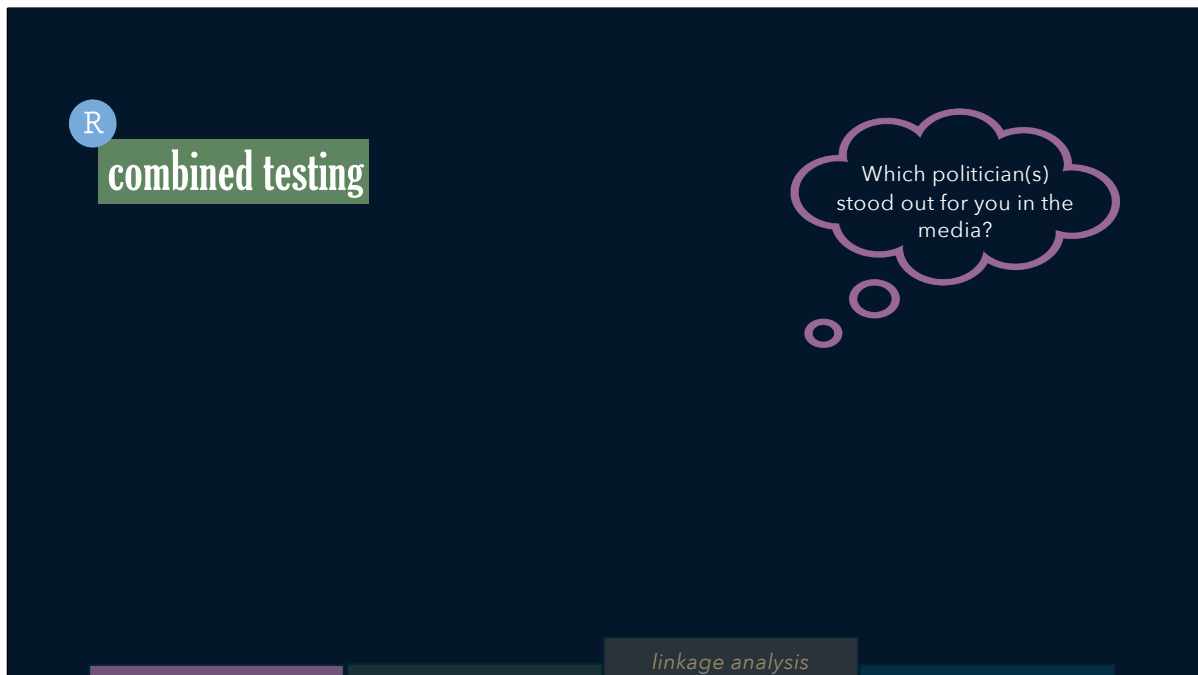
Exactly, we must know what news consumption patterns exist for the respondents in our sample to be able to predict what political coverage they were exposed to!

which (combination of) modalities
in political news coverage
are most influential
for a politician's perceived prominence
by the public?



linkage analysis

Magic!



The question asked in the longitudinal panel was: "which politician(s) stood out for you in the media?"

Respondents could give up to 3 Dutch politicians that stood out for them. The outcome variable was binary and politician-based; 0 meaning the politician did not stand out, 1 meaning the politician did stand out.

Standing out, here, is the dependent variable measuring recall.

R

combined testing



Which politician(s)
stood out for you in the
media?

linkage analysis

This yielded the following results:

1. Merely visual coverage had no positive effect of recalling a politician.
2. Only textual coverage of a politician or their party yielded an odds ratio of 1.06, meaning that with every textual coverage, a politician was 1.06 times more likely to recall the politician as striking in the news.
3. Multimodal coverage, where the party leader was visible in the main image and also referenced by name or party in the title, increased the likelihood of them being recalled with 1.13

So, we conclude that multimodal coverage is a stronger positive predictor than monomodal, that is text-only or visual-only, coverage for politician recall.

Hypothesis confirmed!

R

combined to



h politician(s)
ut for you in the
media?

linkage analysis

Happy hypotheses!

R

combined testing



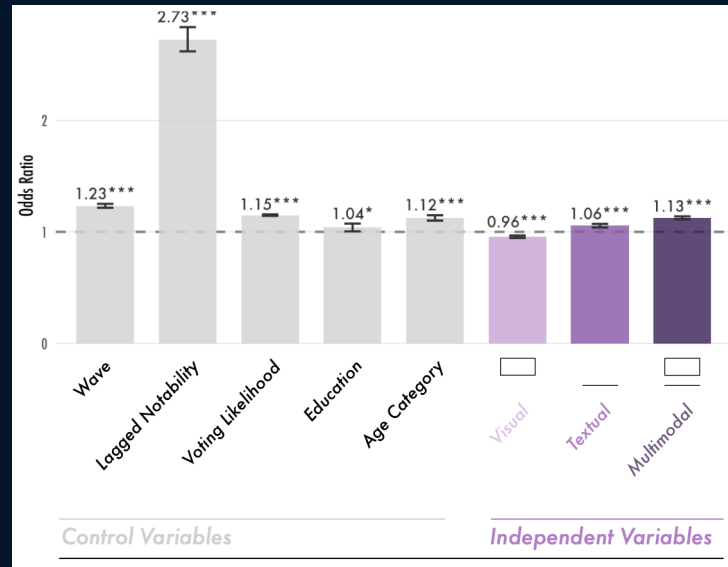
Which politician(s)
stood out for you in the
media?

linkage analysis

Or is there more to the story?

R

combined testing



linkage analysis

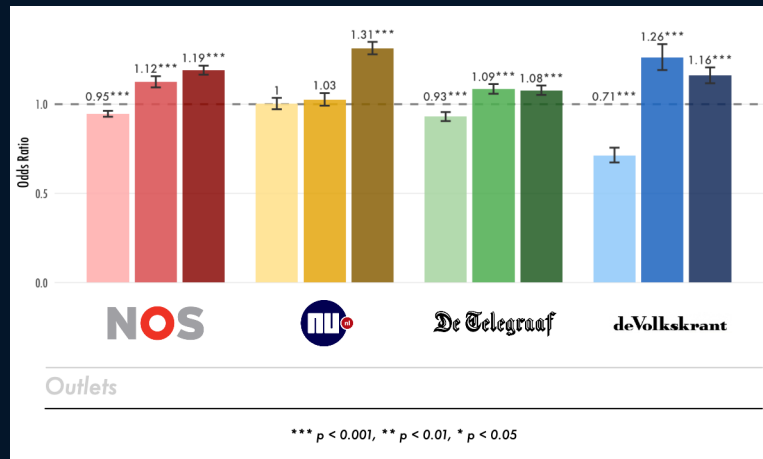
Obviously, there is always more to say. When we include the relevant control variables that we have in our longitudinal panel data, we find that the lagged dependent variable outweighs all other variables in predicting which politician is recalled by a respondent.

This means that when a respondent already tracks a mental library with politicians, either through textual, visual or multimodal memory associations, this respondent draws from that mental library when prompted our question.

Wave, voting likelihood, and age are also strong and sometimes stronger positive predictors of a politician being recalled as standing out in the media.

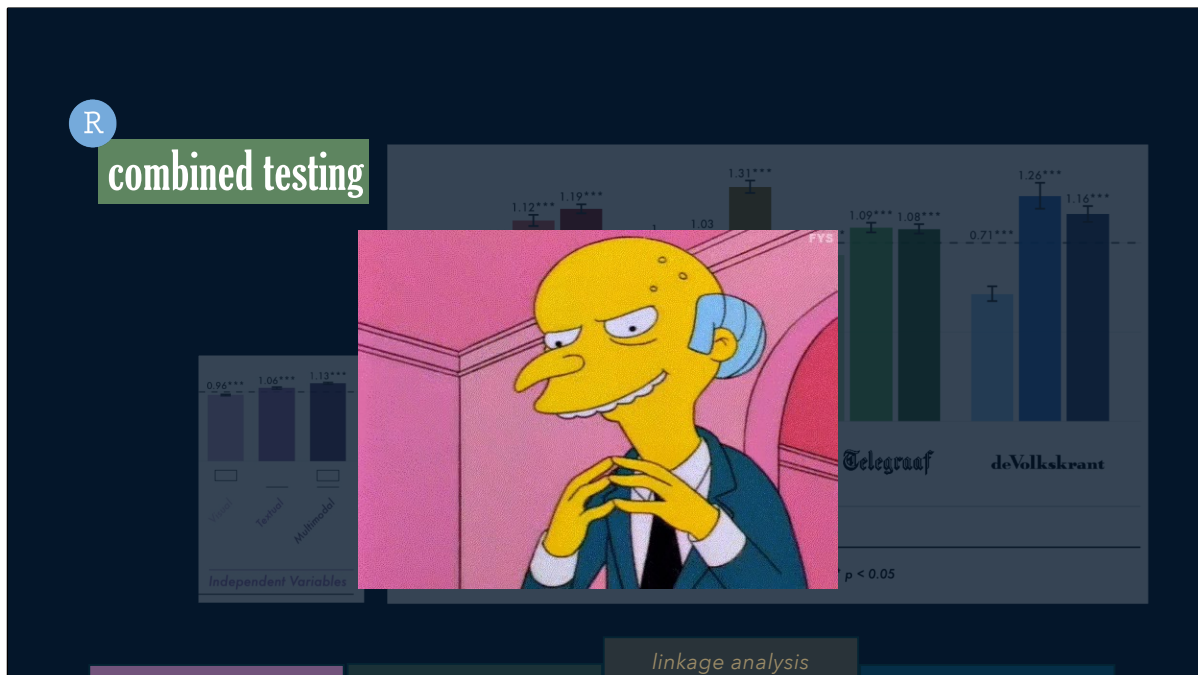
R

combined testing



linkage analysis

Interestingly, we do find different odds ratios for different outlets, and the predictive pattern of multimodal coverage being the strongest positive predictor does not hold for De Telegraaf nor De Volkskrant.



Something that I find particularly interesting and hope to study in another project.

MODALITY MATTERS!!

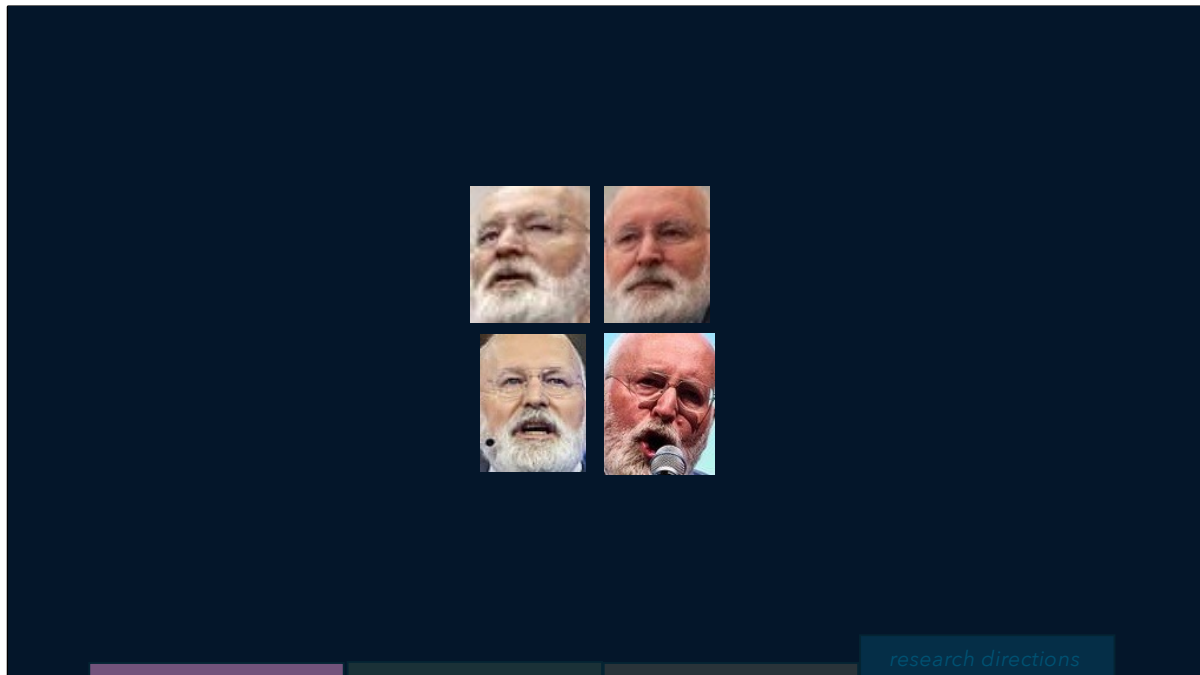
linkage analysis

We carefully conclude that modality matters

MODALITY MATTERS?! *linkage analysis*

But the question is how and in what constellations on the supply and context on the demand side.

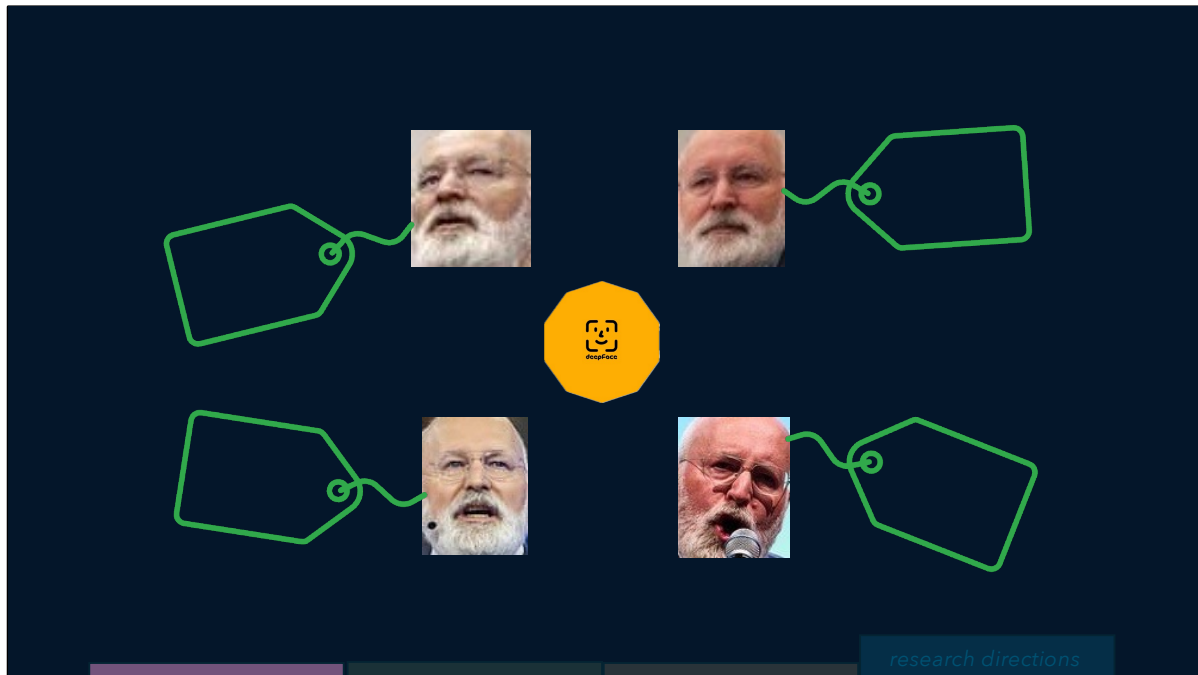
To deepen our understanding of what types of depictions shape a the public's perceptions of these politicians, we must study the multimodal editorial framing further.



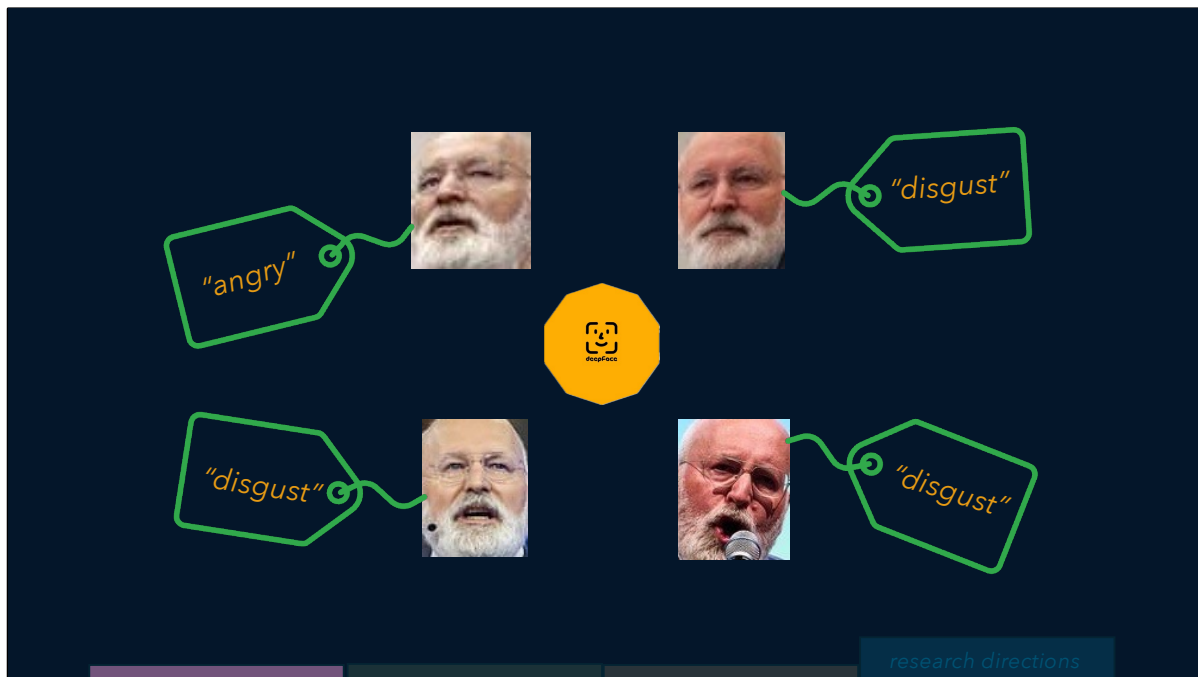
Take the example of the depictions of Frans Timmermans from the examples at the beginning of this presentation.



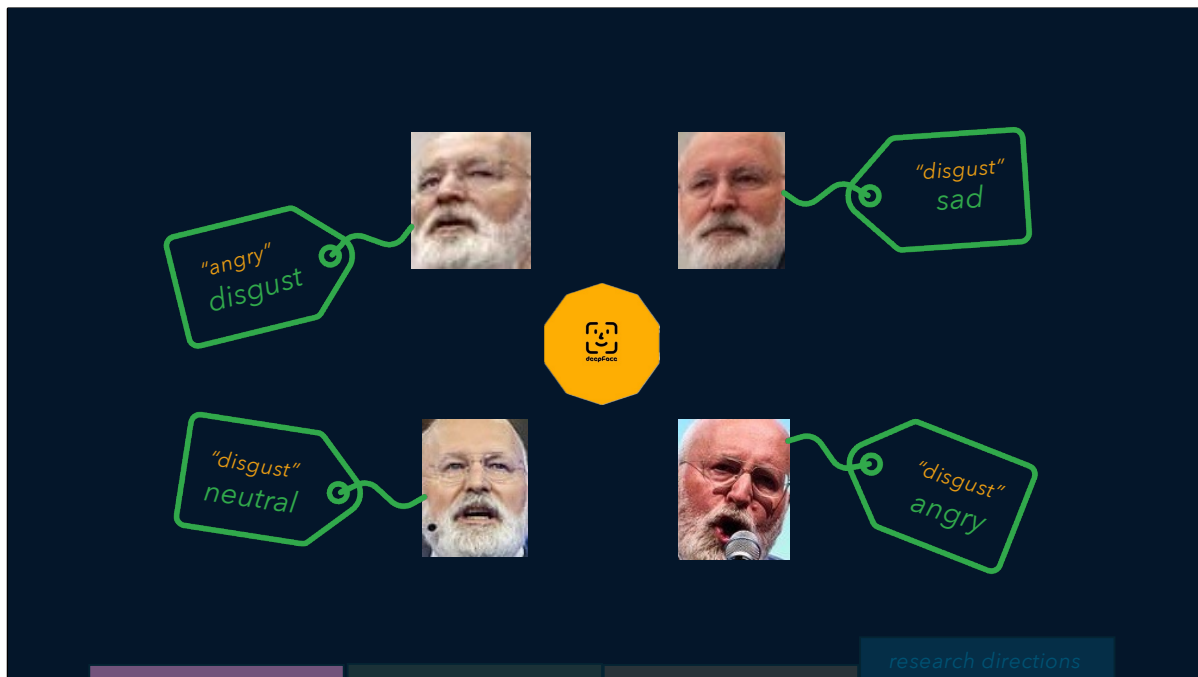
A person can associate the politician with totally different things based on how the politician is depicted.



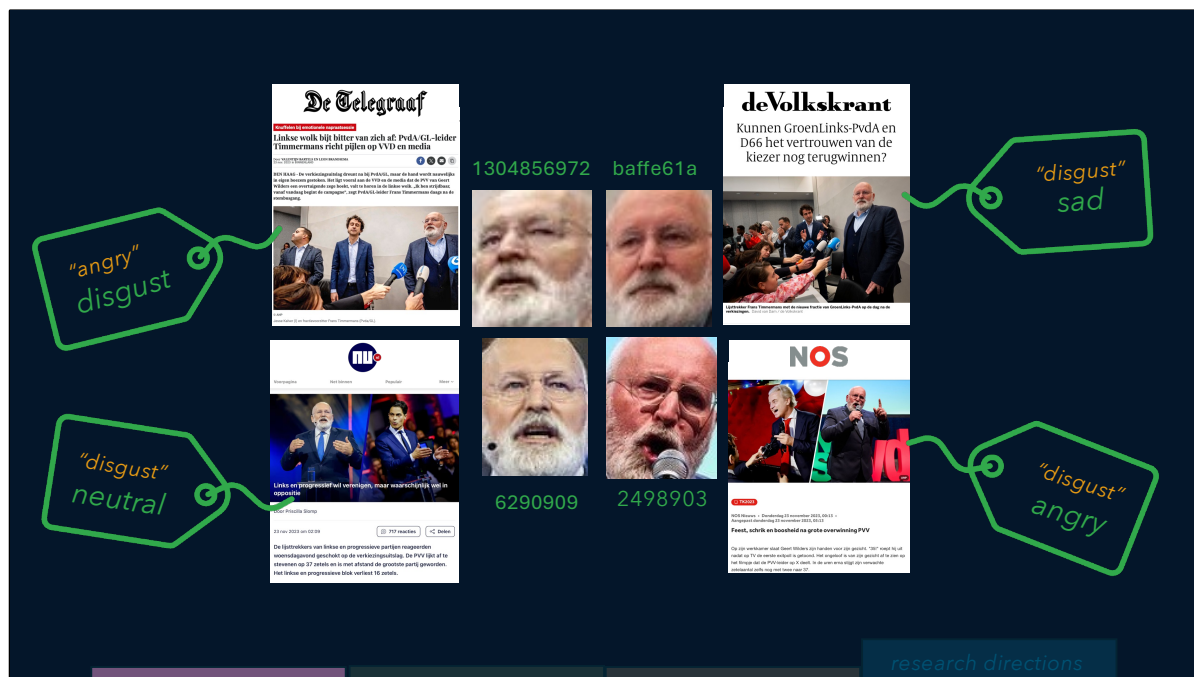
Before the evolution of LLMs, Meta's DeepFace library was considered a reliable source in labelling emotions of facial images. There have been several publications that use this model to classify emotions in politician appearances in online media.



So I tried it on my data. It failed horribly.



On this slide I contrasted the emotions I associated with these depictions of Timmermans in green against the emotions that DeepFace assigned, in orange. As you can see, they rarely overlap.



But I doubt whether it makes sense to guess these so-called emotions without the actual context in the article.

What is the point of studying these depictions in isolation of they are, in reality, always complemented with a textual frame?



WAGENINGEN
UNIVERSITY & RESEARCH

Wies Ruyters, Prof. Dr. Rens Vliegenthart, Prof. Dr. Sanne Kruike-meier, dr. Susan Vermeer



MODALITY MATTERS...!

How Multi-Modal Media Prominence Drives
Political Recognition and Party Support



wies.ruyters@wur.nl

theoretical framework

methodolojourney

linkage analysis

research directions

I am curious to hear your thoughts on this and look forward to a fruitful discussion. Thanks for listening!